Untitled

by Phil

General metrics

23,925	3,490	144	13 min 57 sec	26 min 50 sec
characters	words	sentences	reading time	speaking time

Score

76

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Writing Issues

209	70	139
Issues left	Critical	Advanced

Writing Issues

70	**Correctness**
27	Mixed dialects of english
5	Misspelled words
2	Conjunction use
6	Incorrect noun number
4	Comma misuse within clauses
5	Determiner use (a/an/the/this, etc.)
1	Improper formatting

1	Faulty subject-verb agreement
7	Confused words
10	Wrong or missing prepositions
1	Incorrect phrasing
1	Pronoun use

8 Clarity

8	Wordy sentences

Unique Words

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unique words

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rare words

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5.6

characters per word

Sentence Length

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24.2

words per sentence

Untitled

The integration of technology has had a profound impact on the progress of healthcare sectors. One significant outcome has been the implementation of electronic health records (EHRs), enabling hospitals to efficiently capture, manage, and preserve patient information (Shickel & Tighe, 2017). Moreover, technology has played a vital role in optimising[1] inventory management, healthcare supply management, medicine tracking, and appointment scheduling within healthcare facilities. Ensuring accurate, efficient, and prompt delivery of patient care services holds utmost importance in the healthcare sector. The introduction of EHRs and patient management systems has greatly improved the effectiveness of patient management and healthcare services (Jain et al., 2019). These systems have provided healthcare sectors with the ability to digitally manage their data. However, the complexity of these systems has made manual control challenging, leading to errors and considerable time consumption. Robotic Process Automation (RPA) has emerged as a cutting-edge technology that is shaping the future of various industries (Aguirre, 2017). It falls under the umbrella of the Fourth Industrial Revolution (4IR), alongside Artificial Intelligence, Machine Learning, Sensors, and other innovative technologies. RPA has gained significant traction across sectors such as Business, Production, Education, Healthcare, and more (Ivancic et al., 2019). By combining hardware, software, networking, and automation, RPA simplifies tasks and brings value to core business processes within any industry. RPA has emerged as a solution to streamline and automate healthcare processes, including patient data management, heart monitoring, billing and insurance claims, and Telemedicine (Aguirre, 2017). RPA is an

innovative software. technology that facilitates the development, implementation, and control of software robots or bots that imitate human behaviour[2] while interacting with digital systems and software (Ratia et al.,2018). These software robots can perform tasks such as comprehending on□screen[3] information, executing precise keystrokes, manoeuvring[4] through complex systems, recognising[5], and extracting data, and executing various predefined actions (Van der Aalst, 2018). This technology boasts of quicker and more consistent performance than humans, eliminating the need for breaks or interruptions during operations RPA software robots can perform repetitive and time□consuming[6] tasks accurately and efficiently, freeing up healthcare professionals to focus on providing 12 quality patient care. Hospitals spend 40% of their time on administration, which might be lessened with RPA (Precedence Patient data management is a critical process that is carried out in hospitals to ensure effective healthcare delivery. This process involves a series of steps that include data collection, entry, storage, retrieval, analysis, and sharing (Mercury, 2022). Patient data collection involves completing information such as the patient's name, number, address, age, gender[7] medical aid information. Patient information includes a patient[8] medical record, assigned appointments and[9] visits. All this information is important to manage to ensure privacy. The information can also be used as reference[10] for treatment (University of Pittsburgh, 2022). Additionally[11] patient data management provides an in-depth analysis of patients which will help in decision-making and future trends. The importance of patient data management cannot be overemphasised[12], as it plays a vital role in various aspects of healthcare, such as decision making[13], quality improvement, clinical research, public health, and payment reimbursement (Mercury ,[14]2022). Automation has become increasingly prevalent in the healthcare industry, particularly in developed countries

(Precedence Research, 2022). Developed countries have been able to automate hospital processes, including patient data management, due to their technological advancements and the implementation of RPA. Although there has been a greater emphasis on automating critical processes, such as heart monitoring and life machines, developed countries have also been able to automate the processes involved in patient data management (Juell Skielse et al., 2022). Extensive research has been conducted to explore the integration of RPA in hospitals, and it is evident that developed countries have capitalised[15] on this technology to enhance their healthcare systems (Precedence Research, 2022). The United States and the United Kingdom, for instance, have successfully leveraged RPA to automate critical healthcare processes, resulting in increased efficiency and improved patient experiences. However, it is worth noting that limited information is available regarding the adoption of RPA in clinics. While hospitals have been the primary focus of RPA development and implementation, clinics, which play a crucial role in providing primary and specialised[16] care, seem to have received less attention in terms of RPA utilisation[17] (Dan Brennan, 2021). Further research is warranted to investigate the extent to which clinics have embraced RPA technology. Examining the potential benefits of integrating RPA in clinic settings, such as streamlining appointment scheduling, optimising[18] medical billing, and improving patient data management, would contribute to a more comprehensive understanding of the impact of RPA in the broader healthcare landscape. Hospitals and clinics are both healthcare facilities, but they differ in terms of their scope, size, services provided, and patient population. Here are some key distinctions between hospitals and clinics. Clinics, although they may be considered a subsidiary of hospitals, possess distinctive characteristics that set them apart in terms of scope, size, patient populations, and services (Wang, 2017). 13 While hospitals

typically encompass comprehensive medical services and have larger physical footprints, clinics are more specialised[19] in nature[20] and provide outpatient care, focusing on specific medical disciplines (Brennan et al, 2021). This dissimilarity in scope and size influences[21] the operational processes and workflows within clinics, necessitating tailored RPA implementations to address their unique requirements. Moreover, the patient populations served by clinics differ from those of hospitals. Clinics primarily attend to individuals with less critical health concerns, routine check-ups, chronic condition management, and preventive care. In contrast, hospitals cater to a diverse patient base, including individuals with severe illnesses, injuries, and complex medical conditions (Wang, 2017). Understanding these differences is crucial for designing and implementing RPA solutions that align with the specific needs and workflows of clinics, optimising[22] their operational efficiency and patient care. The current medical technology market in South Africa remains largely reliant on conventional devices, indicating a lag in the adoption of advanced technologies compared to other industries globally. The country, in general, faces challenges in embracing technology, with healthcare being no exception (FTI consulting[23], 2019). The healthcare system in South Africa exhibits a two-tier structure characterised[24] by private[25] and public sector[26]. The private sector, serving only 16% of the population, contributes to over 50% of the total healthcare expenditure, primarily catering to wealthier, urban, and formally employed individuals. This scenario is influenced by the high costs associated with private healthcare and the limited coverage provided by medical aid (mutual health insurance). Conversely, the public sector serves the remaining 84% of the population with fee-free primary care predominantly delivered through nurse-led clinics. However, the public sector faces resource constraints, both in terms of healthcare professionals and infrastructure (FTI consulting[27], 2019). A

considerable portion of public health centers, more than half, continue to rely on paper-based record systems. Internet connectivity, particularly in rural areas, poses a significant challenge, while some regions lack even basic infrastructure such as electricity. Regarding the adoption of EHR technologies, the three major hospital groups, namely Life Healthcare, Netcare, and Mediclinic, which collectively constitute 66% of the private hospital sector, emerge as the primary potential purchasers. These hospital groups represent crucial stakeholders with significant potential for driving EHR implementation in South Africa (FTI consulting, 2019). The adoption of RPA in South Africa's healthcare sector has not been observed due to the country's limited technological infrastructure. Developed countries have extensively studied RPA from their own perspective, but there is no evidence of RPA implementation in South Africa's healthcare industry. RPA in healthcare has been limited in hospitals across South Africa, mainly due to the lack of technological advancements (Cline, 2013). Unlike private hospitals like Netcare and Life Care Healthcare, which have found ways to enhance their healthcare systems technology through the implementation of Electronic Healthcare Records and Electronic Medical Records, there is still a lack 14 of RPA in patient data management processes (Netcare,2022). Although these private hospitals have made strides towards digital transformation, the lack of RPA implementation in patient data management processes still poses a challenge. Patient data management remains a manual and time-consuming process, with healthcare providers having to enter data manually and store it in various formats. With public sector as stated that the public sector serves 84% of the population with fee-free primary care predominantly delivered through nurse-led clinics. These clinics also indicate a lack of RPA implementation some as far as not evening having EHR systems (FTI consulting, 2019). South African clinics are frequently

viewed as a separate entity from South Africa[36] hospitals in healthcare facilities, having its[37] own distinct set of operational procedures (Brennan, 2021). However, the majority of the field's current research has focused on technical developments in hospitals, paying relatively little attention to the possibility for similar developments in clinics. As a result, there is a need to refocus research to explore and advance South African clinics[39] technical capabilities in order to[40] raise the standard of healthcare delivery as a whole. Consequently, the implementation of RPA for patient data management in clinics necessitates a distinct approach when compared to hospitals. The factors mentioned above underscore the significance of evaluating clinics as case studies to examine the application and impact of RPA in their specific context. 1.2. Problem Description and Statement Recent research highlights the absence of technological advancements in healthcare facilities across South Africa, specifically in rural areas and select public healthcare facilities. Within the public sector, a significant constraint remains the lack of adequate infrastructure, with over half of public health centres[41] still relying on paper-based record systems. Nurse-led clinics deliver fee-free primary care to approximately 84% of the population. Even in public hospitals where EHR systems have been implemented, healthcare workers encounter difficulties in using the system efficiently, resulting in reduced time available for patient care. Furthermore, the absence of RPA implementation in both public hospitals and nurse-led clinics exacerbates the time-consuming nature of administrative tasks. As a consequence, patient data management and patient data analysis suffer due to the inability to gather information from paper based[42] records or effectively operate the EHR system for querying and analysing[43] patient data. Although some private sector hospitals have advanced healthcare technology such as Hospital Information Systems and Electronic Health Systems, the

implementation of RPA for patient data management in these systems is yet to be realised (de Villiers, 2020), this indicates that there is still a significant gap in technological advancement in the healthcare sector in South Africa, leading to 15 problems such as manual data entry and increased time spent on paperwork instead of attending to patients. Most research focuses on the implementation of RPA in hospitals, with little mention of clinics, this is due to the focus on large healthcare entities, which include hospitals, as they make up a significant portion of the healthcare industry. Although clinics are healthcare institutions that are dependents of hospitals they should be considered as being hampered by a lack of RPA. Unfortunately, there has been limited research conducted on the implementation of RPA in clinics as the focus is mainly on hospitals. As a result, there is little information available on the adoption of RPA in clinic institutions in South Africa, with most studies centred on technology in major healthcare institutions like hospitals. 1.3. Research Aim and Objectives The aim of this study is to improve existing patient management processes on a clinic system by implementing RPA software. This study aims to provide a solution on how RPA can be effectively utilised to enhance patient data management in a clinic. By achieving the research objectives stated below, this study aims to contribute to the improvement of patient care and offer insight for future efforts to advance the healthcare institutions in South Africa. The research objectives for this study are: • RO1: To evaluate the extent to which RPA has been adopted in patient data management in healthcare institutions. • RO2: To investigate the challenges that come with implementing RPA in healthcare institutions, for patient data management. • R03: To analyse the current patient data management processes in a selected clinic in South Africa and determine which ones are suitable for the implementation of RPA. • R04: To investigate

suitable RPA technologies and methodologies for the implementation of patient data management automation in a South African clinic. • RO5: To implement RPA technology in a South African clinic for the management of patient data. • RO6: To assess the impact of the RPA implementation for patient data management in a South African clinic. 1.4. Research Questions The main research question is How can patient data management processes at a clinic be improved 16 with the implementation of RPA? 1.5. Scope, Constraints and Ethics 1.6. Research Methodology and Chapter Structure Design Science Research (DSR) is a research approach that focuses on creating and evaluating IT artifacts with the aim of solving organisational problems (Gacenga et al., 2012). It is considered a distinct form of science alongside the natural sciences and the human sciences. DSR involves the invention and innovation of new artifacts to address problems or achieve improvements, thereby creating a new reality (Alturki et al., 2013). The methodology of Design Science Research has dominated the fields of Information Systems and Computer Science, it has gained prominence due to its rapid growth. According to Bayazit (2004), Design Science Research Methodology is a systematic inquiry that seeks to understand the configuration, structure, composition, purpose, value, and meaning of man-made things and systems. The fundamental logic of discovery in Design Science Research Methodology is deductive, as it starts with an unsolved problem and seeks to find justificatory knowledge or a kernel theory that can help solve the problem (Piirainen 2014). In Design Science Research, mathematical and computational methods are predominantly used to assess the effectiveness and quality of artifacts. Empirical techniques are also employed to evaluate their performance (Hevner et al., 2004). The value of Design Science Research in the field of Information Systems lies in its application to address business needs in appropriate environments and its

contribution to the knowledge base for further research and practice. The success of an artifact is measured by its ability to solve existing problems, and a theory that lacks practical utility in the environment contributes minimally to the Information Systems literature (Hevner et al., 2004). The Design Science Research Methodology has gained significant traction as a valuable approach in the fields of Information Science and Computer Science. It is highly regarded due to its ability to address human, organizational, and social problems through the development of artifacts (Hevner et al., 2004). The present research project involves the design and development of RPA processes using automation software. The designed processes will be integrated into system[58] at a clinic. To achieve this goal, the Design Science Research Methodology (DSRM) has been identified as the most appropriate research approach. The DSRM is a methodology that emphasises[59] the creation of innovative artifact[60] 17 that contribute to the field of study. In accordance with[61] this methodology, a comprehensive literature review will be conducted to gather useful information and guide the development of the RPA software. The DSRM approach is deemed suitable for this research project because the development of an innovative artifact is a primary objective which will be the implementation of an RPA software in clinic[62,63] system, with the aim of contributing[64] to the healthcare field. In this regard, the research will follow the DSRM's recommended phases, which will be extensively discussed in the dissertation. These phases include the: 1.6.1 Awareness of Problem The Design Science Research Process commences with the identification of the problem to be addressed by the research. In this study, the problem is identified as the lack of RPA implementation in South African clinics to effectively manage patient data. To accurately identify and define the problem, an extensive literature review will be conducted to investigate the current state of RPA implementation in clinic

management systems. The literature review aims to produce a proposal for a new research initiative that will focus on implementing RPA software in a clinic to address the identified problem. This initial phase is of utmost importance as it lays the foundation for the subsequent stages of the DSRM methodology. The success of the overall research project depends largely on the effectiveness of this phase, which will ensure that the problem is well-defined and aligns with the research objectives (Rothenber, 2014). 1.6.2 Suggestion The second phase of the DSRM involves identifying the patient data management processes that need to be automated in a clinic, following the identification of the problem in the first phase. This process will involve conducting interviews and questionnaires with healthcare workers to gain an understanding of the current patient data managed in their system and the processes involved. The information obtained will then be used to identify the patient data management processes that will require automation. There after[65] the next step will involve identifying an appropriate RPA software that can effectively address the automation of the identified processes. To achieve this, various techniques such as literature review, field research and[66] expert opinions will be utilised[67] to generate ideas on the best RPA software to use for the automation of the processes. The identified RPA software solutions will be evaluated based on their capability to automate patient data management processes in the clinic. The features, capabilities, and limitations will be analysed.[68] UiPath, Power Automate, and Blue Prism will be investigated in-depth to determine their suitability. The selected RPA software will need to meet specific criteria to be deemed suitable for addressing the identified problem. This phase is critical as it forms the link between the proposal and the design of the implementation idea. The output of this stage is a set of potential solutions that will be evaluated and 18 refined in the subsequent stages of the DSRM (Keuchler,

2015). 1.6.3 Development In the development phase of the DSRM, the proposed solution is put into practice through the development of an artifact. In the context of this study, the chosen patient data management processes and RPA software are implemented and integrated into the existing clinic patient data management system, to automate the processes. To achieve this, the processes to be automated will be modelled[69] using software tools such as Visio. The implementation of the RPA software onto[70] the clinic system follows, with the installation of the software onto the computers that have the clinic system or external computers that have the system. The modelled[71] RPA processes are then implemented onto[72] the clinic system. Afterward, thorough testing is conducted to evaluate the effectiveness of the RPA software in manipulating the system. Iterative development is then carried out in phases until the final processes are done. The output of this will be an artifact that demonstrates the validity of the proposed solution and its potential to address the identified problem (Keuchler,2015). 1.6.4 Evaluation Once the RPA software has been integrated and implemented into the clinic's system, an evaluation will be conducted to assess the impact of the RPA software on the system. This evaluation will involve a series of instructions that participants will be asked to follow in order to[73] determine whether the automated processes have been successful in performing their intended tasks. Additionally, the evaluation will aim to identify any errors or problems that may have been missed during the development phase, which can then be addressed and resolved in subsequent phases of the methodology. This evaluation is a critical step in determining the effectiveness of the RPA software in improving patient data management processes in the clinic and will inform any necessary adjustments or improvements that may need to be made before the final product is produced (Rothenber, 2014). 1.6.5 Conclusion The conclusion phase of the DSRM involves

assessing the effectiveness of the implemented RPA software and the success of the patient data management processes being automated. This will also involve the documentation of the steps taken during the methodology, any challenges encountered, and the outcomes achieved. The conclusion will also include recommendations for further improvements and future research directions for RPA implementation in clinics and other[74] (Keuchler,2015). 19 1.6.6 Iterative Approach An iterative approach is a fundamental aspect of the Design Science Research (DSR) methodology, which involves a continuous process of designing, developing, evaluating, and refining a solution to a problem. This approach enables the researcher to refine and improve their solutions based on feedback obtained from the evaluation process, which can be used to inform subsequent iterations (Larman, 2004). In line with the DSRM methodology, the implementation of the RPA software involves an iterative process that involves cycles of modelling[75], integration, and testing of the system's processes. Each process is modelled[76], integrated, and tested to determine its effectiveness. If a process does not work effectively, necessary modifications are made until the process achieves its intended goal. The iterative process plays a significant role in the DSR methodology as it offers a flexible and adaptable framework for developing a solution to address complex challenges. Through the iterative refinement and upgrading of the solution, an artefact[77] can develop innovative and efficient solutions that can have a significant impact in[78] the field

1.	~~optimising~~ → optimizing	Mixed dialects of English	Correctness
2.	~~behaviour~~ → behavior	Mixed dialects of English	Correctness
3.	~~on screen~~ → on-screen	Misspelled words	Correctness
4.	~~manoeuvring~~ → maneuvering	Mixed dialects of English	Correctness
5.	~~recognising~~ → recognizing	Mixed dialects of English	Correctness
6.	~~time consuming~~ → time-consuming	Misspelled words	Correctness
7.	and gender	Conjunction use	Correctness
8.	~~patient~~ → patient's	Incorrect noun number	Correctness
9.	, and	Comma misuse within clauses	Correctness
10.	a reference	Determiner use (a/an/the/this, etc.)	Correctness
11.	Additionally,	Comma misuse within clauses	Correctness
12.	~~overemphasised~~ → overemphasized	Mixed dialects of English	Correctness
13.	~~decision making~~ → decision-making	Misspelled words	Correctness
14.	Mercury ,	Improper formatting	Correctness
15.	~~capitalised~~ → capitalized	Mixed dialects of English	Correctness
16.	~~specialised~~ → specialized	Mixed dialects of English	Correctness
17.	~~utilisation~~ → utilization	Mixed dialects of English	Correctness
18.	~~optimising~~ → optimizing	Mixed dialects of English	Correctness
19.	~~specialised~~ → specialized	Mixed dialects of English	Correctness
20.	~~in nature~~	Wordy sentences	Clarity

21.	~~influences~~ → influence	Faulty subject-verb agreement	Correctness
22.	~~optimising~~ → optimizing	Mixed dialects of English	Correctness
23.	~~consulting~~ → Consulting	Confused words	Correctness
24.	~~characterised~~ → characterized	Mixed dialects of English	Correctness
25.	the private	Determiner use (a/an/the/this, etc.)	Correctness
26.	~~sector~~ → sectors	Incorrect noun number	Correctness
27.	~~consulting~~ → Consulting	Confused words	Correctness
28.	~~consulting~~ → Consulting	Confused words	Correctness
29.	~~own~~	Wordy sentences	Clarity
30.	of 14	Wrong or missing prepositions	Correctness
31.	~~towards~~ → toward	Wrong or missing prepositions	Correctness
32.	~~time□consuming~~ → time-consuming	Misspelled words	Correctness
33.	the public	Determiner use (a/an/the/this, etc.)	Correctness
34.	~~evening~~ → even	Incorrect phrasing	Correctness
35.	~~consulting~~ → Consulting	Confused words	Correctness
36.	~~Africa~~ → African	Confused words	Correctness
37.	~~its~~ → their	Pronoun use	Correctness
38.	~~for~~ → of	Wrong or missing prepositions	Correctness
39.	~~clinics~~ → clinic's, clinics'	Incorrect noun number	Correctness
40.	~~in order to~~ → to	Wordy sentences	Clarity

41.	~~centres~~ → centers	Mixed dialects of English	Correctness
42.	~~paper-based~~ → paper-based	Misspelled words	Correctness
43.	~~analysing~~ → analyzing	Mixed dialects of English	Correctness
44.	~~realised~~ → realized	Mixed dialects of English	Correctness
45.	~~dependents~~ → dependent	Incorrect noun number	Correctness
46.	~~of~~ → on	Wrong or missing prepositions	Correctness
47.	~~centred~~ → centered	Mixed dialects of English	Correctness
48.	this study aims	Wordy sentences	Clarity
49.	~~on~~ → in	Wrong or missing prepositions	Correctness
50.	~~clinic~~ → clinical	Confused words	Correctness
51.	~~utilised~~ → utilized	Mixed dialects of English	Correctness
52.	~~for~~ → into	Wrong or missing prepositions	Correctness
53.	~~analyse~~ → analyze	Mixed dialects of English	Correctness
54.	~~for~~ → on	Wrong or missing prepositions	Correctness
55.	, and	Comma misuse within clauses	Correctness
56.	to solve	Wordy sentences	Clarity
57.	~~organisational~~ → organizational	Mixed dialects of English	Correctness
58.	the system	Determiner use (a/an/the/this, etc.)	Correctness
59.	~~emphasises~~ → emphasizes	Mixed dialects of English	Correctness
60.	~~artifact~~ → artifacts	Incorrect noun number	Correctness
61.	~~In accordance with~~ →	Wordy sentences	Clarity

	By, Following, Per, Under		
62.	the clinic	Determiner use (a/an/the/this, etc.)	Correctness
63.	~~clinic~~ → clinical	Confused words	Correctness
64.	to contribute	Wordy sentences	Clarity
65.	~~after~~	Conjunction use	Correctness
66.	, and	Comma misuse within clauses	Correctness
67.	~~utilised~~ → utilized	Mixed dialects of English	Correctness
68.	~~analysed~~ → analyzed	Mixed dialects of English	Correctness
69.	~~modelled~~ → modeled	Mixed dialects of English	Correctness
70.	~~onto~~ → into	Wrong or missing prepositions	Correctness
71.	~~modelled~~ → modeled	Mixed dialects of English	Correctness
72.	~~onto~~ → into	Wrong or missing prepositions	Correctness
73.	~~in order to~~ → to	Wordy sentences	Clarity
74.	~~other~~ → others	Incorrect noun number	Correctness
75.	~~modelling~~ → modeling	Mixed dialects of English	Correctness
76.	~~modelled~~ → modeled	Mixed dialects of English	Correctness
77.	~~artefact~~ → artifact	Mixed dialects of English	Correctness
78.	~~in~~ → on	Wrong or missing prepositions	Correctness